

# Aviya Litman

AI4SCIENCE, COMPUTATIONAL GENOMICS, DEEP LEARNING, PRECISION MEDICINE, NEUROGENOMICS

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## Education

### Princeton University

PH.D. IN QUANTITATIVE AND COMPUTATIONAL BIOLOGY

- Advisor: Professor Olga Troyanskaya
- Thesis: *AI-driven decomposition of genetic, phenotypic, and clinical heterogeneity*

Princeton, NJ

2022 - Present

### Princeton University

MASTERS IN QUANTITATIVE AND COMPUTATIONAL BIOLOGY

Princeton, NJ

2022 - 2024

### Columbia University, Columbia College

BACHELORS IN MATHEMATICS

- Cumulative GPA: 3.88

New York, NY

2018 - 2021

## Experience

### Troyanskaya Lab, Princeton University

PHD CANDIDATE

September 2022 - Present

- Developed a scalable framework to apply unsupervised mixture modeling for data-driven decomposition of phenotypic and clinical heterogeneity. Enabled the first robust, replicable classification of autism into distinct clinical subclasses with differing genetic and molecular underpinnings (accepted at **Nature Genetics**) [GitHub]
  - Work featured in *The Transmitter* (Spectrum) [press coverage]
- Developed **Otari**, the first sequence-based graph deep learning model for predicting transcriptome-wide, isoform-level regulatory, expression, and variant effect changes across 30 human tissues and brain regions. Trained on genomic sequence and long-read transcriptomic data, Otari revealed widespread variant-driven isoform dysregulation across large regulatory disease mutation datasets, including an autism cohort (submitted to *Nature Computational Science*) [GitHub]
- Leveraged sequence-based deep learning to characterize RNA binding protein dysregulation in a large chronic kidney disease cohort, highlighting its potential role in disease pathogenesis (abstract published in the *Journal of the American Society for Nephrology*)
- Advanced and engineered core features of **Selene**, a Python library and command line interface for training deep neural networks from biological sequence data [GitHub]

Princeton, NJ

### Flatiron Institute, Simons Foundation

SUMMER RESEARCH ASSOCIATE

Summers 2024 and 2025

- Led key research efforts with the Center for Computational Biology and the Simons Foundation Autism Research Initiative (SFARI), advancing deep learning and statistical genomics approaches for studying autism. Work laid the foundation for ongoing doctoral projects.

New York, NY

### Korem Lab, Columbia University Irving Medical Center

DSI RESEARCH SCHOLAR / RESEARCH ASSOCIATE

October 2020 - August 2022

- Advisor: Prof. Tal Korem
- Awarded competitive research funding by the Columbia Data Science Institute and selected as a Data Science Institute Scholar.
- Evaluated and applied **CoPTR**, a bioinformatics tool for analyzing temporal growth dynamics in the microbiome. Demonstrated its accuracy in predicting growth in natural bacterial populations, outperforming prior state-of-the-art methods (published in *Genome Research*).
- Developed **Copangraph**, a data structure, analytic framework, and suite of algorithms for modeling extensive genomic variability across diverse metagenomes (manuscript in preparation).

New York, NY

## Publications

1. **Aviya Litman**, Natalie Sauerwald, LeeAnne Green Snyder, Jennifer Foss-Feig, Christopher Y. Park, Yun Hao, Ilan Dinstein, Chandra L. Theesfeld, and Olga G. Troyanskaya. *Decomposition of phenotypic heterogeneity in autism reveals underlying genetic programs*. *Nature Genetics*, accepted (2025). [preprint]
2. **Aviya Litman**, Zhicheng Pan, Joyce Fang, Ksenia Sokolova, Tess Marvin, Natalie Sauerwald, Christopher Y. Park, Chandra L. Theesfeld, and Olga G. Troyanskaya. *Variant-resolved prediction of context-specific isoform variation with a graph-based attention model*. Submitted to *Nature Computational Science* (2025).
3. Tyler A. Joseph, Philippe Chlenski, **Aviya Litman**, Tal Korem, and Itsik Pe'er. *Accurate and robust inference*

of microbial growth dynamics from metagenomic sequencing reveals personalized growth rates. *Genome Research* (2022) 32(3):558-568. [\[link\]](#)

4. Tess Marvin, **Aviya Litman**, Chen Wang, Chandra Theesfeld, Laura Mariani, Krzysztof Kiryluk, Matthias Kretzler, Olga Troyanskaya, and CureGN Genetics. *Characterizing the Genetic Architecture of Rare Glomerulonephropathy Disease: The Landscape of RNA Binding Protein Dysregulation*. *Journal of the American Society of Nephrology* (2023) 34(11S):p 416. [\[link\]](#)
5. Izaak Coleman, **Aviya Litman**, Ayushi Sharma, and Tal Korem. *Modeling large-scale metagenomic variability with Copangraph uncovers structural variation associated with host health*. *Manuscript in preparation* (2025).

## Selected Talks

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1. Princeton Precision Health Lecture Series, invited speaker. Princeton, NJ, March 2025
2. Winter Q-Bio Conference, main talk. Honolulu, Hawaii, February 2025
3. Lewis-Sigler Institute Symposium, main talk. Trenton, NJ, October 2024
4. Computational Systems for Integrative Genomics Conference, poster presentation. New York, NY, July 2024
5. NHGRI Research Training and Career Development Annual Meeting, poster presentation. Seattle, Washington, April 2024
6. American Society of Nephrology Conference, poster presentation. Philadelphia, Pennsylvania, November 2023

## Awards

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- 2024 **NSF GRFP Honorable Mention**
- 2022 **Lewis-Sigler Institute Scholars Award**, Princeton University
- 2020-21 **Columbia Data Science Institute Scholar**, Research Fellowship (\$10k)
- 2019 **Columbia Emerging Scholar in CS**
- 2019 **Columbia Engineering Fast Pitch Competition Finalist**
- 2018-21 **Dean's List for Academic Excellence**, all semesters

## Outreach

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### Computational Systems for Integrative Genomics Conference

*Flatiron Institute, New York, NY*

CONFERENCE ORGANIZER

*July 2024*

- Led organization of a multi-day research conference including speaker recruitment, abstract review, scheduling, activities, and participant outreach for over 80 attendees from **21 domestic and international institutions**.

### REU in Computational Biology for Previously Incarcerated Students

*Princeton, NJ*

VOLUNTEER INSTRUCTOR

*June 2023*

- Prepared and taught lessons on Python programming, data analysis, and machine learning as part of the REU program for previously incarcerated individuals at Princeton University [\[GitHub\]](#).

### Women in Computational Biology @ Princeton

*Princeton, NJ*

ORGANIZER

*May 2023 - Present*

- Organizing lunch discussions among female-identifying students working in computational biology, and between students and faculty members in the field.
- Building a local community and fostering connections.

### Columbia Undergraduate Admissions / Visitors Center

*New York, NY*

OUTREACH INTERN

*May 2019 - August 2019*

- Prepared and delivered presentations and STEM-themed campus tours to prospective students and their families.

## Skills

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|-------------------------|-----------------------------------------------------|
| <b>Programming</b>      | Python, C/C++, Cython, MySQL, HTML/CSS, LaTeX, Bash |
| <b>Data science</b>     | Numpy, Scipy, Pandas, Scikit-learn                  |
| <b>Machine learning</b> | PyTorch, Lightning, wandb                           |
| <b>General</b>          | Linux/UNIX, Git, Vim, Slurm / Cluster environment   |
| <b>Languages</b>        | English, Hebrew                                     |